

Computational Fluid Dynamics (CFD) Modeling of Bubble Dynamics in the Aluminum Smelting Process

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ABSTRACT: This paper presents a microscale modeling approach for investigation of bubble dynamics in the aluminum smelting process. The motion of a single bubble has been studied through a computational fluid dynamics (CFD) model facilitated with the volume-of-fluid (VOF) method to capture the bubble shapes. Using a two-dimensional geometry of part of a real cell as the testing bed, the motion of different sized bubbles has been simulated in an air–water system and a CO₂–cryolite system. Comparisons between the two systems are conducted through the three periods of bubble motion: bubble sliding under the anode, bubble releasing at the anode edge, and bubble rising in the side channel. It was found that both systems show similar trends in bubble dynamics, such as an increase in the bubble sliding velocity as the bubble size increases and the appearance of a thick head at large bubble sizes. Quantitatively, there are differences between the two systems, evidenced in terms of the detailed bubble dynamics at each period of bubble motion, such as the bubble morphology, the bubble sliding velocity, the bubble layer thickness, and the bubble-induced liquid flow. The detailed microscale modeling provides useful information for the development of a multiscale modeling methodology by building constitutive correlations to support the macro/process scale modeling.

1. INTRODUCTION

The Hall–Héroult process is the only commercial process for producing aluminum from alumina.¹ In an aluminum reduction cell, alumina is fed to, and dissolved in, a molten bath of cryolite at approximately 970 °C in which several carbon anodes are submerged. Electric current is fed between the anodes and an underlying cathode to cause electrochemical reduction of the alumina reactant to aluminum which settles onto a pool lying over the cathode. CO₂ gas bubbles are generated by the reaction at the anode, which causes recirculation flows as a result of the gas bubbles moving up through the molten cryolite (the bath) under the influence of buoyancy. Because cryolite will dissolve most potential wall materials, a layer of frozen cryolite must be formed on the walls of the vessel to contain the bath, and this requires the achievement of a delicate heat balance in the cell, over which the recirculatory flows in the bath have an important influence.

The gases are generated at the bottom surface of the anodes in a continuous manner. Thus, the anode bottom surfaces are covered with a layer of bubbles right beneath the anode bottom surface. The bubble area coverage can vary from 30 to 90%,^{2–4} which leads to an extra voltage drop. According to Haupin,⁵ the extra voltage drop in the electrolyte due to the presence of gas bubbles is in the range 0.15–0.35 V. The bubble motion beneath the anodes also introduces waves into the bath–metal interface, voltage fluctuations, and high local current density, and indirectly results in instabilities of the magnetic field. Moreover, the global scale bath flow and alumina mixing are closely related to the bubble behavior. Therefore, a detailed understanding of the bubble dynamics and the resulting bath flow is important to quantitatively assess its effect on cell performance.

The hostile environment (high temperature and corrosive molten salt bath) restricts direct observation of bubble behavior in industrial cells. Studies of bubble behavior in industrial cells, laboratory cells, and physical models have been reviewed by Cooksey et al.⁶ There is good evidence that the bubble layer thickness is at least 5 mm in industrial cells,⁵ and similar in laboratory cells.^{6–8} In order to observe the bubble dynamic behavior at a scale typical of industrial cell geometries, room temperature laboratory models have been used.^{9–28} Transparent materials such as *Plexiglas* are used to construct the cell, and a room temperature liquid is used to replace the cryolite bath. As listed in Table 1, various gas–liquid systems have been used to represent the CO₂–cryolite system, such as NaOH solution,⁹ CuSO₄ solution,¹⁰ air–oil–water,¹¹ or simply air–water.^{12–28} Since the bubble formation is quite complex and the motion is controlled by many factors, such as surface tension, contact angle, anode shape, and even the roughness of the surface, none of those systems can closely match all the factors of the real system. The standpoint for using the air–water system is that the kinematic viscosity of water is very similar to that of cryolite ($1.005 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$ for water and $1.43 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$ for cryolite). This will lead to a similar liquid flow dynamics as long as the same volume of gas is used but might not have relevance on the similarity of bubble dynamics.

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Table 1. Analogous Systems to CO₂–Cryolite Reported in the Literature

bath liquid	gas	metal	references
NaOH solution	oxygen	none	ref 9
Bluestone solution	oxygen	none	ref 10
light mineral oil	air	water	ref 11
water	air	none	ref 12
			ref 13
			ref 14
			ref 15
			ref 16
			ref 17
			ref 18
			refs 24–27
			ref 28
oil	air	none	ref 14
water–0.5 pct butanol	air	none	ref 15
propaneol–water	air	none	ref 19
isopropanol–water	air	none	ref 19
glycerine–water	air	none	ref 19
olive oil	air	none	ref 20
water	air	organic	ref 21

According to a mathematical simulation,^{22,23} the bubble morphology mainly depends on the liquid's Morton number (Mo), a dimensionless number defined as $((g\mu_l^4(\rho_l - \rho_g))/(\rho_l^2\sigma^3))$, where g is the gravitational acceleration, μ is the viscosity, ρ is the density, and σ is the surface tension. The subscripts g and l stand for the bubble gas and the surrounding liquid, respectively. As shown in Table 2, the Morton number for the air–water system is very different from that for the CO₂–cryolite system. The Eötvös number $((g(\rho_l - \rho_g)L^2)/\sigma)$ is often used together with the Morton number to characterize the shape of bubbles or drops moving in a surrounding fluid or continuous phase. L refers to the characteristic length, e.g., bubble size or diameter (d_b), in this calculation. The Eötvös numbers are provided in Table 2 as well. Because the two systems have a similar ratio of the buoyancy force to the surface tension (i.e., 1.36×10^5 for an air–water system and 1.56×10^5 for a CO₂–cryolite system), consequently, the Eötvös number of the air–water system is very similar to that of the CO₂–cryolite system at a given bubble size.

Over the last two decades, with advances in computing speed, parallelization technology, improved software, and multiphase algorithms, computational fluid dynamics (CFD) has progressed substantially. The advantage of CFD modeling is not only that it provides a cost-effective way to gain a detailed understanding of the complex process but also that it is sometimes the only research tool due to measurement

challenges in a real process, such as the high temperature and corrosive environment found in aluminum electrolytic cells.

Depending on the application and information required, gas–liquid flows such as those encountered in aluminum reduction cells can be modeled at different length scales: at the micro or individual bubble scale or at the macro level by local averaging. The former approach tracks the interfaces around each bubble, and detailed transient bubbling behavior can be obtained. The micro model is very useful to the detailed elucidation of the governing mechanisms of these complex processes but requires large computing power, which means that it cannot presently be directly applied to industrial-scale simulations. The macro model represents the flow field averaged over time, and hence, steady state equations are often solved. The model requires less computing power, but the accuracy of a macro model depends on the accuracy of constitutive correlations that describe the local averaging of the micro scale information, such as bubble-induced turbulence and the turbulent dispersion force. In the past, physical models have often been used to validate the CFD model either based on flow patterns²⁹ or detailed liquid velocity and turbulence³⁰ obtained using particle image velocimetry (PIV) technology.³¹ The validated CFD model is then extended to simulate the exact scale, properties, and operating conditions of the real system.^{32,33} The question lies in whether the CFD model, validated based on the air–water system, can be directly used for another system or not. Using a multiscale modeling approach, the detailed micro modeling information of a real system, such as the bubble dynamics in an aluminum reduction cell, can be used to build constitutive correlations to improve macro modeling accuracy. This multiscale modeling approach is believed to be promising and powerful, and has received increasing interest in the study of complex multiphase flow systems.^{34–39}

The individual bubble model has recently been used to study an aluminum smelting system with different focuses. Einarsrud¹¹ studied the effect of detaching bubbles on aluminum–cryolite interfaces; Das et al.²⁰ investigated the principal characteristics of the detachment and sliding mechanism of gas bubbles under an inclined anode; Wang and Zhang⁴⁰ studied the effect of the anode edge on bubble release. In our own recent work,⁴¹ the relationships between the air–water system and CO₂–cryolite systems have been checked. The assessment was focused on the bubble dynamics during the bubble sliding process under the anode only.

This paper aims to further evaluate the relative effect of bubble sizes and simulation systems (air–water vs CO₂–cryolite) during the bubble releasing process at the anode edge and the bubble rising process in the side channels. In addition

Table 2. Physical Properties of the CO₂–Cryolite and the Air–Water Systems

properties		CO ₂ at 960 °C	cryolite at 960 °C	air at 25 °C	water at 25 °C
density (kg/m ³)		0.4	2100	1.225	998.2
dynamic viscosity (kg/m·s)		1.37×10^{-5}	3.0×10^{-3}	1.789×10^{-5}	1.003×10^{-3}
kinematic viscosity (m ² /s)		3.43×10^{-5}	1.43×10^{-6}	1.46×10^{-5}	1.005×10^{-6}
surface tension coefficient (N/m)		0.132		0.072	
contact angle (deg)		120°		60°	
Morton number		1.645×10^{-10}		2.664×10^{-11}	
Eötvös number at different bubble sizes	0.0113 m	19.904		17.294	
	0.0226 m	79.626		69.175	
	0.0339 m	158.624		137.820	

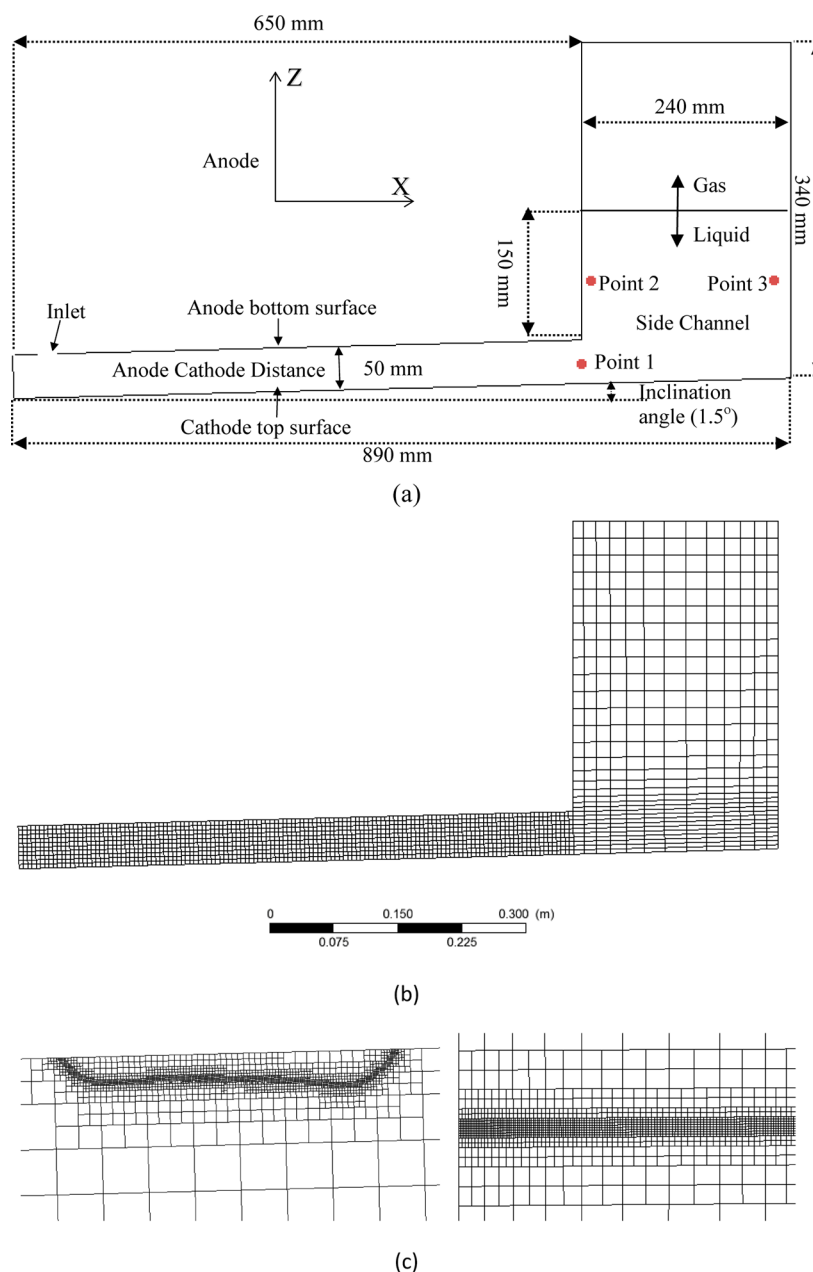


Figure 1. Geometry and mesh information for the CFD model: (a) geometry; (b) initial meshes; (c) refined meshes around the bubble (left) and air–liquid interfaces using the mesh adaption method.

to the study of bubble dynamics, the bubble-induced liquid motion is quantified as well. The investigation is based on similar simulation conditions to those used in our previous study.⁴¹ Also, we identified some errors in the data processing in our earlier work,⁴¹ so we repeated the simulation in the present study with an optimized gas injection method. Hence, the present work amends and extends the research presented in our previous publication,⁴¹ and provides updated information and further discussion.

2. CFD MODELING METHOD

2.1. Model Description. Simulation of bubble flow phenomena requires accounting for the irregular deformation of the interfaces and free surfaces, which has been a challenging task in the numerical modeling of multiphase flow over the past few decades.⁴² Generally, a free surface flow can be modeled by

the interface-tracking method⁴³ or interface-capturing method.⁴⁴ The interface-tracking method uses interface-fitted moving grids, while the interface-capturing method uses fixed grids and solves an additional equation to locate the free surface. Given that the free surface can change its topology due to bubble breakage, overturning, and splashing, the grid might not be able to deform to such an extent. The simulation domain would need to be remeshed, which is computationally intensive. The interface-capturing method is commonly used. There have been many kinds of interface-capturing methods developed, such as the level set,⁴⁵ height of liquid,⁴⁶ and volume of fluid (VOF) methods.⁴⁷ In addition to the traditional CFD approach, the lattice Boltzmann (LB) model has been increasingly used to study interface flow due to its capability to model interfaces.⁴⁸

In this study, the VOF method in the CFD software (ANSYS/Fluent) was used. Details of this method are well documented in the literature.⁴⁹ Here, only the key governing equations are briefly provided.

The VOF method is based on the fact that two or more fluids are not interpenetrating. For each phase, a variable is introduced to record the volume fraction of the phase in each computational cell or control volume. The volume fractions of all phases sum to unity at each cell. The fields for all variables and properties are shared by the phases and represent volume-averaged values, as long as the volume fraction of each of the phases is known at each location. Thus, the variables and properties in any given cell are either purely representative of one of the phases or representative of a mixture of the phases, depending on the volume fraction values. Each phase agrees with the mass conservation equation, given by

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \vec{v}) = 0 \quad (1)$$

where α is the phase volume fraction and t and $\vec{v}$ represent the time and velocity, respectively. In Fluent, the equation will not be solved for the primary phase (the liquid phase in current model setting). The primary phase volume fraction will be computed on the basis of the following constraint:

$$\alpha_g + \alpha_l = 1 \quad (2)$$

As all variables and properties are shared by the phases and represent volume-averaged values, a single momentum equation is solved, with the following formulation:

$$\begin{aligned} \frac{\partial(\rho \vec{v})}{\partial t} + \nabla \cdot (\rho \vec{v} \vec{v}) \\ = -\nabla P + \nabla \cdot (\mu(\nabla \vec{v} + \nabla \vec{v}^T)) + \rho \vec{g} + \vec{F} \end{aligned} \quad (3)$$

where P is the pressure term and $\vec{F}$ represents the surface tension force. The superscript T denotes the matrix transpose operation. The mixture density and viscosity is based on the volume averaging, given by

$$\rho = \alpha_g \rho_g + \alpha_l \rho_l \quad (4)$$

$$\mu = \alpha_g \mu_g + \alpha_l \mu_l \quad (5)$$

The simulation accuracy is largely dependent on the method of interpolation near the interfaces, which affects the convection and diffusion fluxes through the control volume faces in eqs 1 and 3 as well as the surface tension force ($\vec{F}$) that is included as a body force in eq 3. The geometric reconstruction scheme and a mesh adaption method are adopted to capture the detailed interface shapes.

2.2. Simulation Conditions. While there are no technical difficulties in simulating the three-dimensional (3D) motion of bubbles, the limitations of simulation time and computing cost mean that the investigations are conducted using a two-dimensional (2D) geometry. The 2D study may not fully represent the real bubble behavior, which is indeed three-dimensional, but will give a similar trend to the 3D study. The 2D geometry might be acceptable for comparison purposes as long as the air–water and CO₂–cryolite systems have the same simulation conditions.

Figure 1a shows the simulation geometry that represents a slice of a typical commercial Hall–Héroult prebake cell; note

that the geometry is not related to any specific cell design. The distance between anode and cathode is 50 mm. The space or gap between the anode and cathode is conventionally called the anode–cathode distance (ACD) in the aluminum smelting industry. The space between the anode and the side vertical wall is called the “side channel”, which is set at 240 mm in width. An inclination of 1.5° (as might occur because of anode consumption) is set to help the release of bubbles. The formation of bubbles in an aluminum reduction cell is very complex, and the detailed bubble formation mechanism is not fully understood. It is believed that different sized bubbles exist in the cell. The large bubbles move faster than the small bubbles. The bubbles will coalesce to form larger bubbles when the large bubbles catch up with the small bubbles. The bubbles will break up when the bubble size grows too large. To model the bubble generation more realistically, a CFD model fully coupled with thermal and electric models with chemical reactions is required. In this study, a simple gas generation method is used. The gas is injected through the inlet that is located at the anode bottom surface. The inlet is 5 mm in length with the center 25 mm from the left side. The inlet velocity is 0.25 ms^{−1}, and the injection time is set to 0.08, 0.32, and 0.64 s to form initial bubbles with equivalent diameters of 11.3, 22.6, and 31.9 mm, respectively. These three sizes are used on the basis of experimental observations.^{12,18,28} The bubbles undergo various types of distortion under the anode and break up in the side channel. For convenience of discussion, the bubble size refers to the initial introduced bubble size, e.g., the equivalent diameter of the 2D bubble. Following the injection of the specified amount of gas, the inlet boundary condition reverts to a nonslip wall boundary condition. The simulation cases are presented in Table 3.

Table 3. Simulation Cases with Different Bubble Sizes

	system	equivalent bubble diameter (m)
case 1	air–water	0.0113
case 2	CO ₂ –cryolite	0.0113
case 3	air–water	0.0226
case 4	CO ₂ –cryolite	0.0226
case 5	air–water	0.0339
case 6	CO ₂ –cryolite	0.0339

To capture the bubble surface, the mesh size should be substantially smaller than the bubble size. The solution-adaptive mesh refinement feature of ANSYS/Fluent allows the user to refine and/or coarsen meshes based on geometric and numerical solution data. Figure 1b shows the initial mesh without refinement functions, which consists of 1624 quadrilateral cells. During the simulation, the hanging node adaption method⁴⁹ is used, which allows more accurate capturing of the detailed interface between gas and liquid, and significantly reduces computing time. The mesh is dynamically refined at the interface, and coarsened back when the interface moves out of that region. The left part of Figure 1c shows the meshes over a bubble in the bath under the anode, and the right part shows the meshes at the gas–bath interface after mesh adaption at an instance of the simulation. The maximum cell surface is set as 5×10^{-6} m², and the minimum cell surface is set as 6.0×10^{-9} m². The initial time step is set to 5×10^{-5} s. The ANSYS Fluent solver will refine the time step based on the input for the maximum Courant number. The Courant number is a dimensionless number that expresses the ratio of the simulation

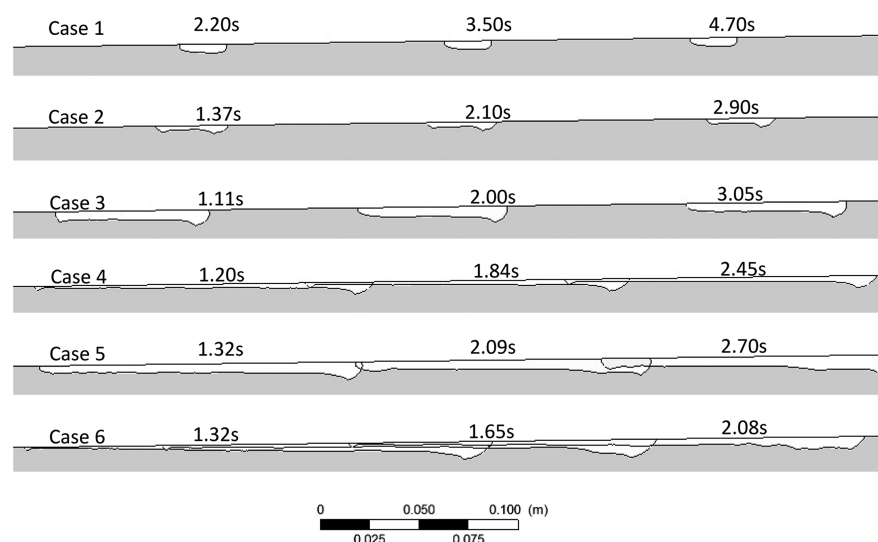


Figure 2. Bubble morphologies in ACD for simulation cases 1–6 at three different times after their dynamically stable states are reached. Case 1: air–water system, bubble size 11.3 mm. Case 2: CO₂–cryolite system, bubble size 11.3 mm. Case 3: air–water system, bubble size 22.6 mm. Case 4: CO₂–cryolite system, bubble size 22.6 mm. Case 5: air–water system, bubble size 33.9 mm. Case 6: CO₂–cryolite system, bubble size 33.9 mm. All cases are plotted from the same location: the left edge is 0.136 m from the left side of the full simulation domain, and the length is 0.445 m. (Adapted from ref 41. Copyright 2012 The Minerals, Metals & Materials Society. Used with permission.)

time step to the characteristic time of transit of a fluid element across a control volume. The global Courant number is set as 0.15 in the current simulations. The absolute convergence criteria for continuity and velocity are both set as 10^{-7} . Given that the liquid flow driven by a single bubble is not strong and that the mesh size is fine enough to capture the local recirculation near the bubble region, the flow is treated as laminar flow.

3. RESULTS AND DISCUSSION

Bubbles demonstrate different behavior in different regions of the cell. In general, the bubble motion can be divided into three periods: bubble sliding under the anode, bubble releasing at the anode edge, and bubble rising in the side channel. The three periods are discussed separately.

3.1. Bubble Sliding under the Anode. **3.1.1. Bubble Morphology.** Following gas injection through the inlet, a single bubble starts to grow. Once the bubble is sufficiently large, it begins to move along the anode bottom surface toward the higher end of the anode. This sliding motion occurs for the three selected bubble sizes. The induced bubble rapidly reaches a dynamically stable state and moves toward the channel between the anode and the cell wall with various shapes which depend on the bubble size and system. Figure 2 shows the bubble morphologies for each case, all of which are after the bubble reaches a dynamically stable state at three instants in time. To be comparable, all cases are plotted at the same scale and in the same location: the left edge is 0.136 m from the left side of the full simulation domain, and the horizontal length is 0.445 m.

When the bubble size is very small, the specific surface area, defined as the total surface area divided by the bubble volume, is large, which implies that a unit volume of gas experience larger surface tension for the smaller bubbles compared to larger bubbles. The bubble tends to a more circular form as the bubble size decreases.

The formation of bubble shapes is actually a complex dynamic balance of surface tension, buoyancy force, viscosity

friction between liquid and gas, solid wall friction, and contact angle. For all of the simulation cases, the bubble becomes flat where the thickness is less than its length. Thus, bubble shape is quite stable, as different time frames show a similar shape. When the bubble size increases, the bubble simply becomes more flattened, and the bubble tends to form a nonregular shape with a thick head, and a long and thin tail. This trend becomes more obvious as the bubble size increases. Such bubbles with thicker head and thinner tail represent a midsection of a large bubble moving under a downward facing surface and have been well observed in the past studies either in aluminum reduction cells^{12,19,22,24–28} or more broadly in gravity–current or density–current flow behavior.⁵⁰

3.1.2. Bubble Layer Thickness. Figure 3 shows the mean bubble thickness for all simulation cases. For both systems, the

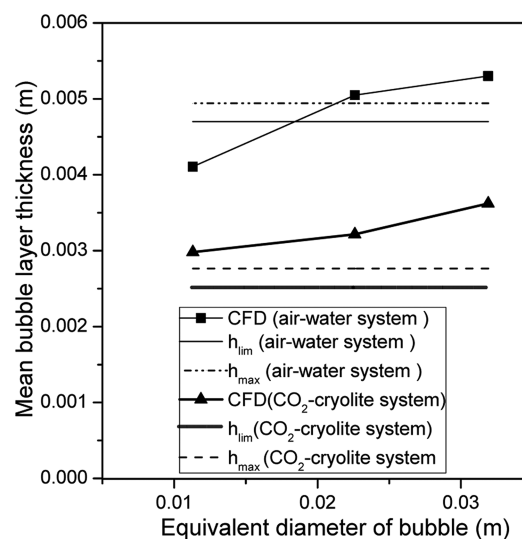


Figure 3. Mean bubble thickness as a function of bubble diameter and system. (Adapted from ref 41. Copyright 2012 The Minerals, Metals & Materials Society. Used with permission.)

bubble thickness increases as the bubble diameter increases. At a given bubble diameter, the bubble thickness is larger in the air–water system ($\sim 4\text{--}5\text{ mm}$) than that in the CO_2 –cryolite system ($\sim 3\text{ mm}$). It is interesting to see that the predicted bubble thickness based on the single bubble model is quite consistent with experimental work with continuous gas injection.⁵

The bubble thickness below flat surfaces has been studied in the past.^{51–53} For a bubble layer under a solid surface, its thickness increases with increasing bubble volume until it reaches a maximum value, then subsequently increases in the volume and slightly decreases the bubble depth until it reaches the limit height. The empirical formulas for the maximum and limiting heights of a stationary bubble under a downward surface were given as follows:⁵¹

$$h_{\max} = (-0.19 \times (\pi - \theta)^2 + 1.293 \times (\pi - \theta) - 0.053) \times \sqrt{\frac{\sigma}{(\rho_1 - \rho_g)g}} \quad (6)$$

$$h_{\lim} = \sqrt{2 \times (1 - \cos(\pi - \theta))} \times \sqrt{\frac{\sigma}{(\rho_1 - \rho_g)g}} \quad (7)$$

In the above equation, θ is the contact angle.

Given that the surface inclination angle in this work is very small, the work on flat surfaces is still valid for a comparison with the simulation results. The above equation has been actually used to compare the measured bubble thickness under a slightly inclined surface relevant to aluminum cells.¹⁸ According to eqs 6 and 7, the maximum and limiting heights of CO_2 bubbles in molten cryolite are 2.8 and 2.5 mm, respectively, and the maximum and limiting height of air bubbles in water are 4.9 and 4.7 mm, respectively.

As shown in Figure 3, the simulated bubble thicknesses are in a reasonable range of the empirical correlations developed for bubbles below a flat surface. Closer agreement is not possible, as the simulation demonstrates an effect of bubble size, while the empirical correlations do not consider the bubble size effect. The simulated bubble thickness from the air–water system is closer to the empirical correlations than the CO_2 –cryolite system. Possibly, the empirical correlation is based on the air–water data, and is not directly applicable for the CO_2 –cryolite system. In eqs 6 and 7, the term $(\sigma/((\rho_1 - \rho_g)g))^{1/2}$ on the right-hand side (RHS) gives a similar value for the two systems, while the coefficients of this term in each case are functions of the contact angle. This implies that the contact angle plays a dominant role in determining the difference in bubble thickness between the two investigated systems. Figure 4 shows the values of these coefficients at a wide range of contact angles. As indicated by the slope of the curves, the effect of contact angle increases as the contact angle increases.

3.1.3. Bubble Sliding Velocity and Drag Coefficient. The motion of bubbles is quantified by their mean velocity at their dynamically stable state. Figure 5 shows the mean bubble velocity as a function of the bubble size for both systems. The mean velocity increases as the bubble size increases for both systems. When the bubble size increases, the buoyancy force increases linearly with the bubble volume, while the bubble surface area does not increase proportionally. Thus, the bubble receives less skin friction per volume of gas for larger bubbles; consequently, larger bubbles slide faster than smaller bubbles.

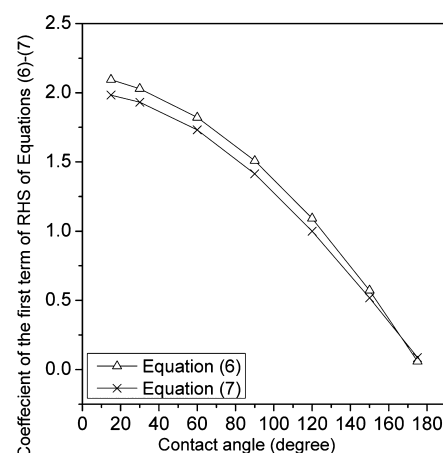


Figure 4. Effect of contact angle on the coefficient of the term on the right-hand side of eqs 6 and 7.

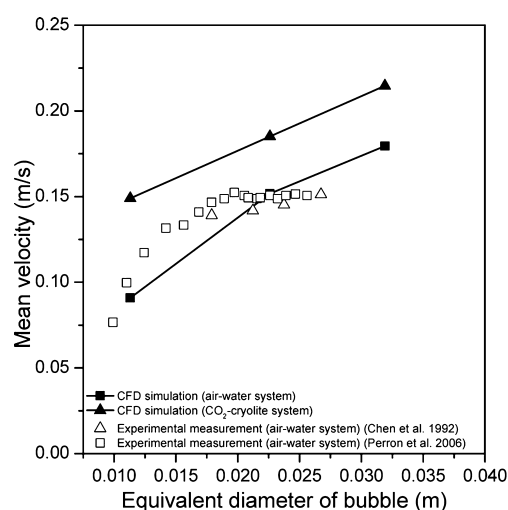


Figure 5. Mean sliding velocity as a function of bubble diameter and system. (Adapted from ref 41. Copyright 2012 The Minerals, Metals & Materials Society. Used with permission.)

For a given bubble size, the bubble velocity is higher for the CO_2 –cryolite system compared to the air–water system. This appears consistent with the bubble shapes: as the bubble thickness is smaller in the CO_2 –cryolite system, consequently there is less resistance around the bubble head.

The sliding velocity under an inclined channel has been measured experimentally in the past.^{18,24–28} While quantitative comparison between the CFD simulation and the experimental measurement is not possible, as they are conducted at different conditions, it is important to know if the CFD prediction is in the range of the physical modeling or not. The experimental data from two previous works involving air–water systems are used for this comparison.^{18,27} As shown in Figure 5, the CFD simulation is within a reasonable range of the experimental data. The measurements from both Che et al.²⁷ and Perron et al.¹⁸ were conducted at an inclination angle of 2.0° , while the 2D CFD simulation is at an angle of 1.5° . Further quantitative comparison is necessary by implementing a CFD model in 3D and at the same inclination angle as the experiment.

The bubble-induced voltage drop is closely related to bubble coverage area and gas layer thickness. The bubble coverage area will have a much larger effect than the bubble thickness.⁶ Figure

3 indicates that the bubble coverage area is larger in the CO₂–cryolite system than in the air–water system for the same size bubble. On the other hand, the larger bubble thicknesses in the air–water system often lead to lower bubble sliding velocity; thus, the bubble residence time under the anode is larger in the air–water system than in the CO₂–cryolite system for the same size bubble. Therefore, a combined consideration of bubble coverage area and bubble velocity is needed to compare the effect on the voltage drop between the two systems.

A drag coefficient is often used to quantify the drag or resistance of an object in a fluid environment, where a lower drag coefficient indicates the object will have less hydrodynamic drag. The bubble is often treated as a 3D sphere in the empirical correlations. Thus, the drag coefficient is calculated on the basis of a form for spherical bubbles, given as

$$C_d = \frac{4}{3} \frac{(\rho_l - \rho_g) d_{bg}}{\rho_l v^2} \sin(\varphi) \quad (8)$$

where C_d is the bubble drag coefficient and φ is the inclination angle of the anode bottom surface.

Figure 6 shows the drag coefficient of the bubble as a function of the bubble size for both systems. For a given bubble

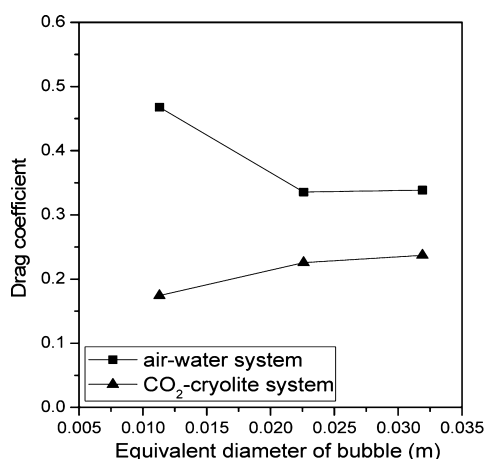


Figure 6. Bubble drag coefficient as a function of bubble size for different systems. (Adapted from ref 41. Copyright 2012 The Minerals, Metals & Materials Society. Used with permission.)

size, the drag coefficient in the air–water system is larger than that in the CO₂–cryolite system. The drag coefficient does not change much for the two large bubbles. It is interesting to observe that the drag coefficient increases for the air–water system and reduces for the CO₂–cryolite system at the smallest bubble size. It is commonly accepted that the drag coefficient will increase as the bubble size reduces due to the large volume-to-surface ratio. As shown in Figure 2, at small sizes (cases 1 and 2), the bubble undergoes a lot more distortion in the CO₂–cryolite system than it does in the air–water system. This may explain why the drag coefficient from the CO₂–cryolite at the small size (case 2) does not follow the trend of the air–water system (case 1). Further investigation, with more simulations of different bubble sizes, is required to better explain this issue.

According to Ishii and Zuber's correlation,⁵⁴ the drag coefficient is about 2.67 in the free rising bubbling region. The drag force coefficient related to flow under an inclined surface is almost 10 times less than that in free rising bubbling

flow in the CO₂–cryolite system. A proper set of drag coefficients is required in the local averaged model for process scale simulation, where the drag coefficient determines the interface momentum exchange. In previous studies, using the local averaged model, the drag coefficient under the anode is simply set the same as if it were vertical flow,⁵⁵ or arbitrarily set with a small value.^{29,32,33} Present quantification of the drag coefficient for bubbles under the anode indicates the capability of the current modeling as a useful approach to further evaluate the drag coefficient in the region between the anode and cathode for improving the accuracy of process-scale simulations.

3.1.4. Flow Dynamics around and within the Bubble. In an aluminum electrolytic cell, the alumina is mainly consumed in the region between the anode and cathode, and a good transfer of alumina from the surrounding bath into this region is required to ensure the cell operates with good performance. The bubble-induced liquid flow is believed to be the main driver for alumina mixing within the ACD. It is interesting to check the bubble-induced liquid flow within this region by plotting the flow dynamics around the bubbles. While the single bubble model cannot fully represent the real condition, it provides an advantage to assess the contribution of each individual bubble.

Figure 7 plots the streamlines that show the liquid flow around the bubble. In this figure, all cases are located with the left edge 0.136 m from the left side of the full simulation domain and with a length of 0.275 m. The buoyancy component induced by the density difference between the two phases, which runs parallel to the anode's bottom surface, is the main cause of the bubble sliding along the anode, while the perpendicular component of liquid flow flattens the shape of the bubble. The bubble head pushes the front liquid away, and the liquid flows backward under the bubble to fill the space where the bubble tail had been. Right beneath the bubble's bottom surface, the liquid flows forward driven by the skin friction at the gas–liquid interface. This leads to various local recirculations. The local recirculation increases as the bubble size increases due to the large momentum exchange between the liquid and the gas bubble.

Figure 8 plots the flow velocities and dynamic pressure distribution in the bubble region. To keep the same scale in all of the figures, only the bubble head region is plotted for cases 3–6. Note that the spatial positions are not the same in this figure. The velocity magnitude inside the bubble is much larger than that in the surrounding liquid. Thus, the forward motion of gas will be impeded by the front liquid. This will lead to a momentum exchange at the interface. Consequently, there is a high dynamic pressure gradient right in front of the bubble head. The pressure gradient partly drives the liquid forward and partly drives the liquid downward, leading to a thick head. As the bubble size increases, the gas bubble velocities increase, leading to a higher dynamic pressure gradient. At the gas–liquid interface, the skin friction leads to higher liquid flow to the front. This forward flow meets with the backward flow beneath the bubble head, resulting in the formation of a big bubble nose. Negative dynamic pressures are observed in the regions associated with local swirls. The detailed flow dynamics explain the formation of different types of bubbles very well.

3.2. Bubble Releasing at the Anode Edge. Figure 9 shows the bubble behavior at the anode edge for all simulation cases. When the bubble approaches the edge of the anode, the horizontal momentum keeps the bubble moving along. Once

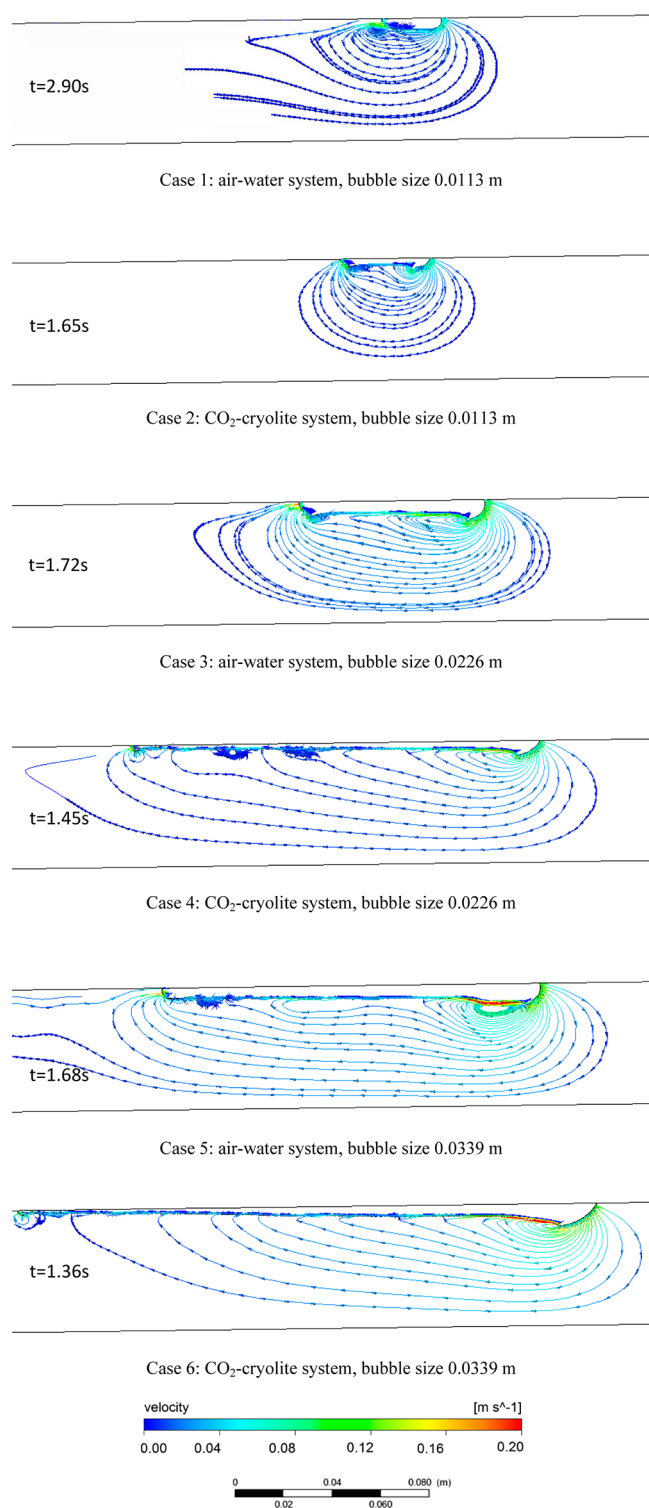


Figure 7. Streamlines showing liquid flow around the bubble for simulation cases 1–6.

the bubble escapes the constraint of the anode bottom surface, it changes direction and rises upward under the strong effect of the buoyancy force.

For smaller bubbles (Figure 9a,b), the bubble tail closely follows the bubble head and releases as a whole with little gas left beneath the anode. For larger bubbles (Figure 9c–f), the bubble breaks into several parts. The bubble head rises up and leaves the tail section beneath the anode. This will be because

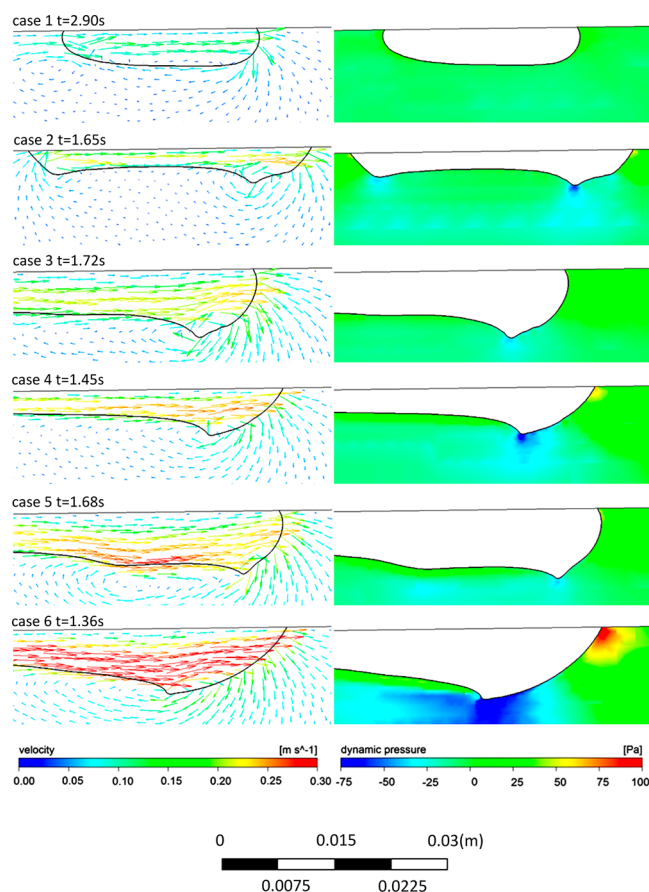


Figure 8. Flow dynamics around the region of the whole bubble (cases 1 and 2) or the bubble head (cases 3–6) expressed by velocity vectors (left) and dynamic pressure (right). Case 1: air–water system, bubble size 0.0113 m. Case 2: CO₂–cryolite system, bubble size 0.0113 m. Case 3: air–water system, bubble size 0.0226 m. Case 4: CO₂–cryolite system, bubble size 0.0226 m. Case 5: air–water system, bubble size 0.0339 m. Case 6: CO₂–cryolite system, bubble size 0.0339 m.

the buoyancy force of the bubble head and the frictional force of the sliding tail are significantly more than the strength of the bubble which will be dependent on the surface tension, viscosity, bubble size, etc. Even though there is a fair amount of gas left beneath the anode bottom surface for the large bubbles, the released part is still much larger for the large bubbles than for the small bubbles.

In comparison between the two systems, it is interesting to see that the rising bubbles detach from the anode vertical wall in the air–water system (Figure 9a,c,e), while they closely attach to the anode vertical wall in the CO₂–cryolite system (Figure 9b,d,f). It is apparent that this difference is largely due to the very different contact angles in these two systems.

Figures 10 (case 3) and 11 (case 4) show the velocity vectors for four time frames during the bubble release process for each system when the bubble size is 22.6 mm. The bubbles exhibit a violent motion with much larger velocities inside the bubble. Various local patterns of recirculation can be observed, and the strong shear at the gas–liquid interface eventually leads to bubble breakup and formation of small bubbles (Figures 10d and 11d).

3.3. Bubble Rising in the Side Channel. Undoubtedly, the bubble release pattern around the edge of the anode directly influences the subsequent bubble behavior.

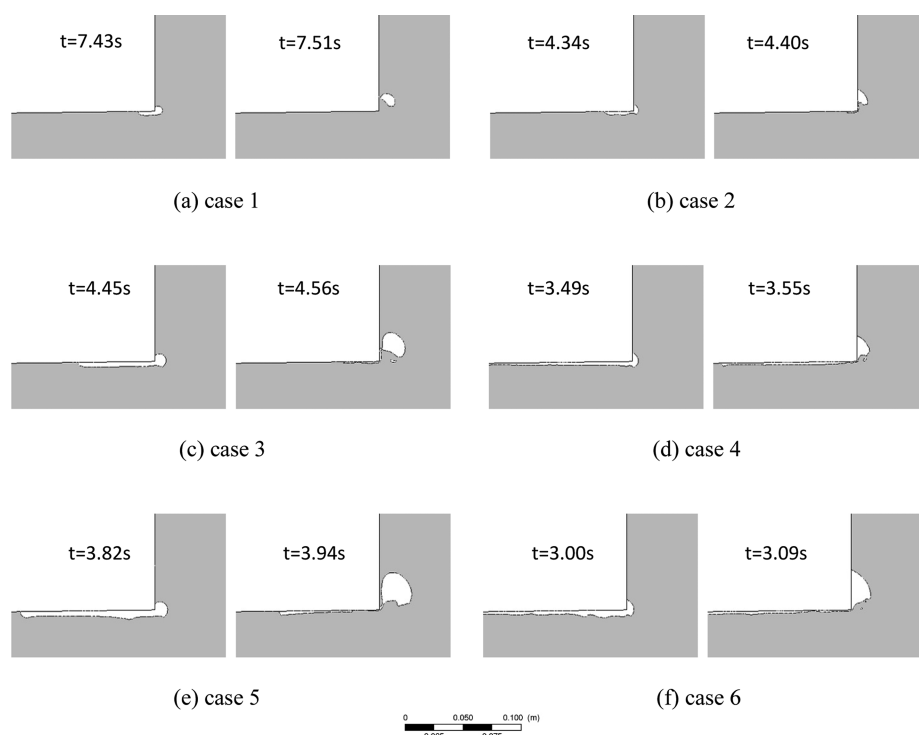


Figure 9. Snapshots showing the bubble escaping process at the anode edge: (a) air–water system, bubble size 0.0113 m; (b) CO_2 –cryolite system, bubble size 0.0113 m; (c) air–water system, bubble size 0.0226 m; (d) CO_2 –cryolite system, bubble size 0.0226 m; (e) air–water system, bubble size 0.0339 m; (f) CO_2 –cryolite system, bubble size 0.0339 m.

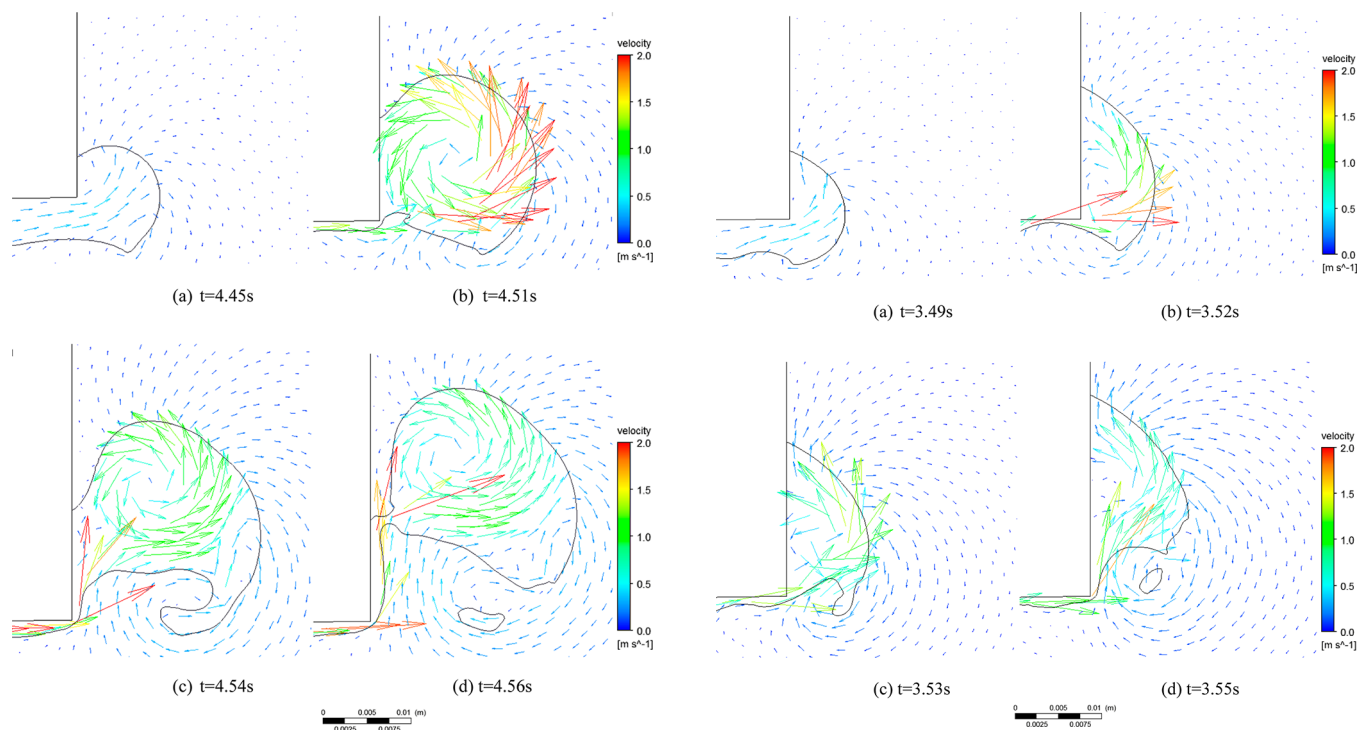


Figure 10. Velocity vectors for four time frames of simulation case 3 (air–water system, bubble size 0.0226 m).

Figure 11. Velocity vectors for four time frames of simulation case 4 (CO_2 –cryolite system, bubble size 0.0226 m).

Figure 12 plots the bubble morphologies in the side channel for each case at three different times. As might be expected, small bubbles do not break up during the vertical motion. Larger bubbles tend to split into smaller bubbles and those smaller bubbles closely follow the parent body for some time.

The distinction of bubble release behavior for the two systems is continued in the side channel. For the air–water system, the bubble completely detaches from the anode and quickly moves away. Conversely, for the CO_2 –cryolite system, the bubble

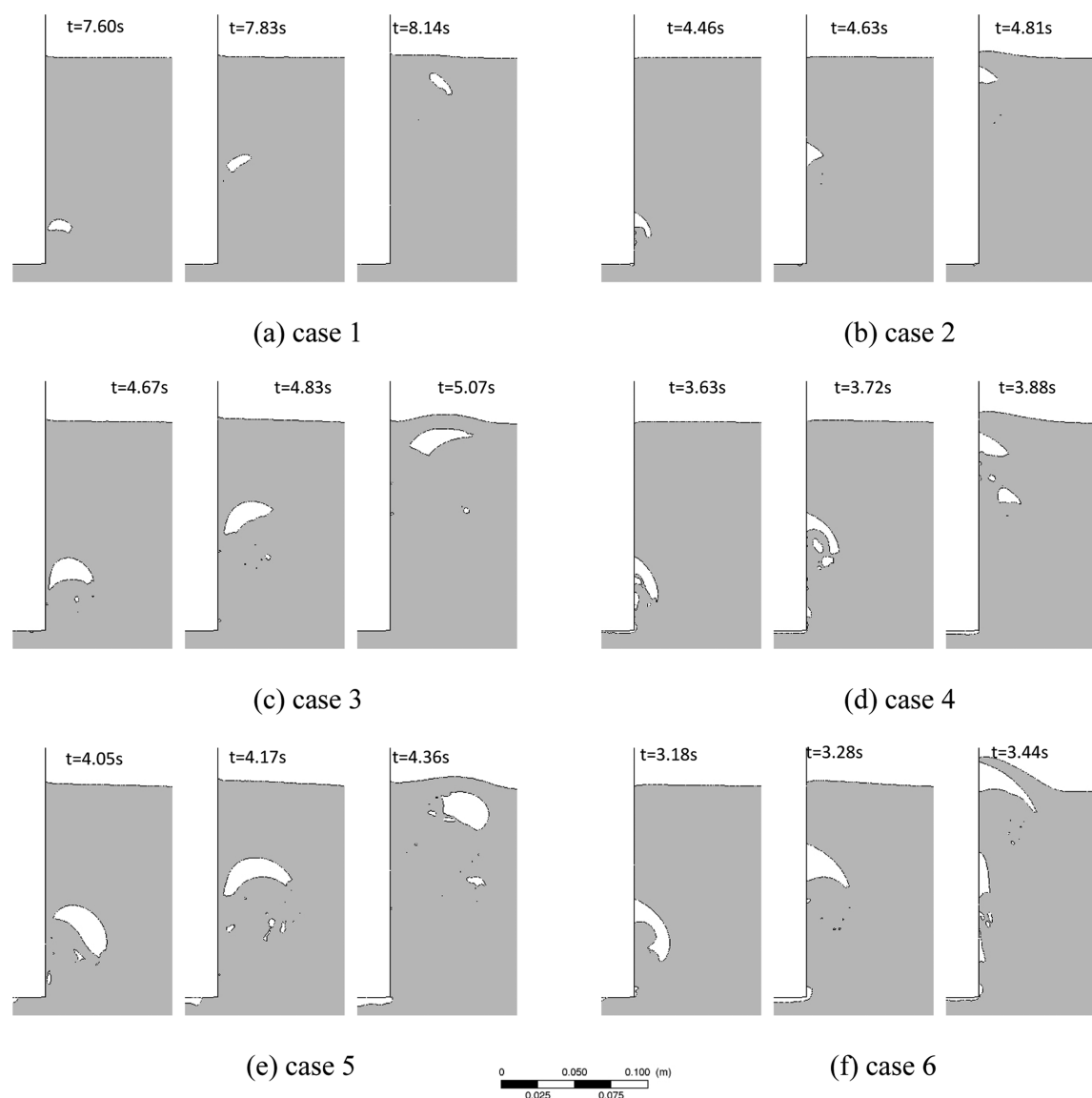


Figure 12. Bubble morphologies for different simulation cases at three different times in the side channel of the cell: (a) air–water system, bubble size 0.0113 m; (b) CO₂–cryolite system, bubble size 0.0113 m; (c) air–water system, bubble size 0.0226 m; (d) CO₂–cryolite system, bubble size 0.0226 m; (e) air–water system, bubble size 0.0339 m; (f) CO₂–cryolite system, bubble size 0.0339 m.

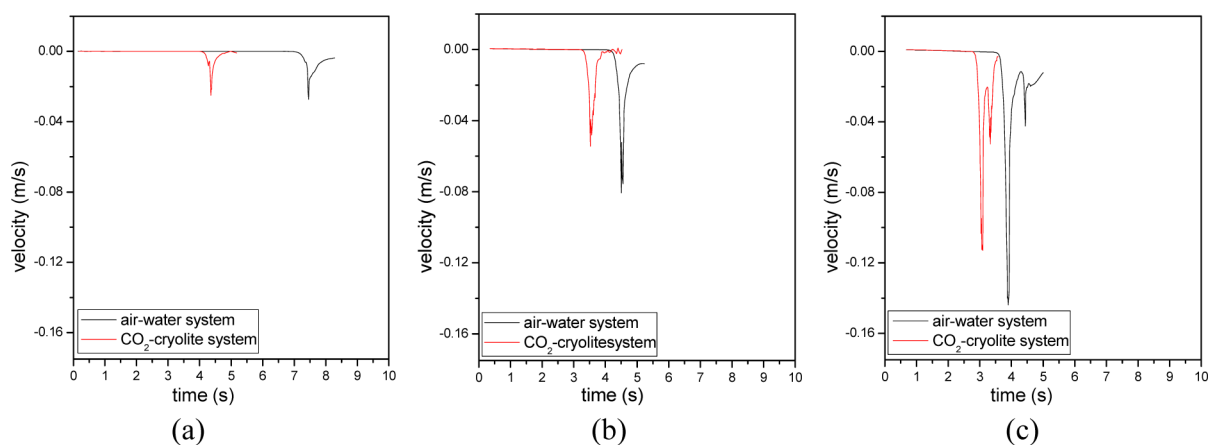


Figure 13. Liquid horizontal velocity at point 1 for each case: (a) bubble size 0.0113 m; (b) bubble size 0.0226 m; (c) bubble size 0.0339 m.

keeps moving along the anode edge all the way through the side channel.

The bubbles are the main driver for the liquid flow. It is interesting to examine the relative effect of simulation systems

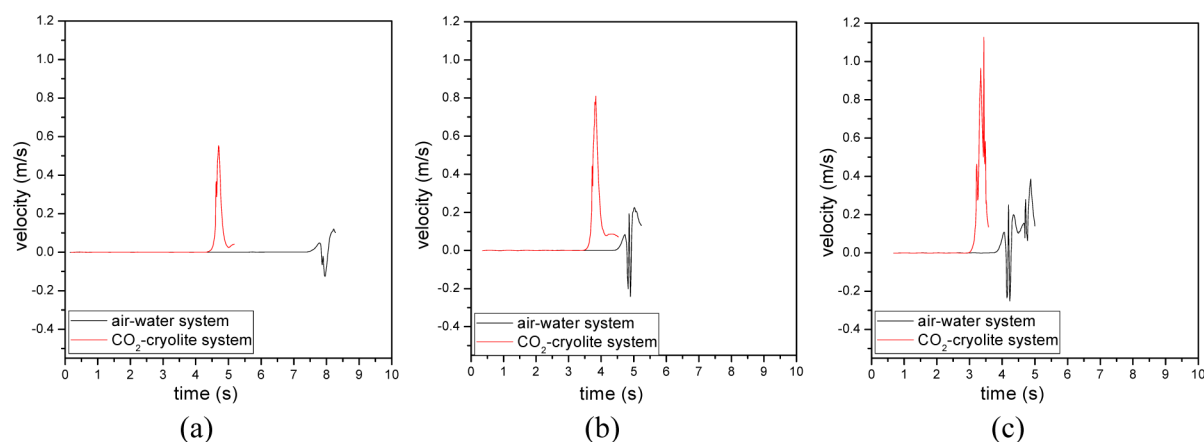


Figure 14. Liquid vertical velocity at point 2 for each case: (a) bubble size 0.0113 m; (b) bubble size 0.0226 m; (c) bubble size 0.0339 m.

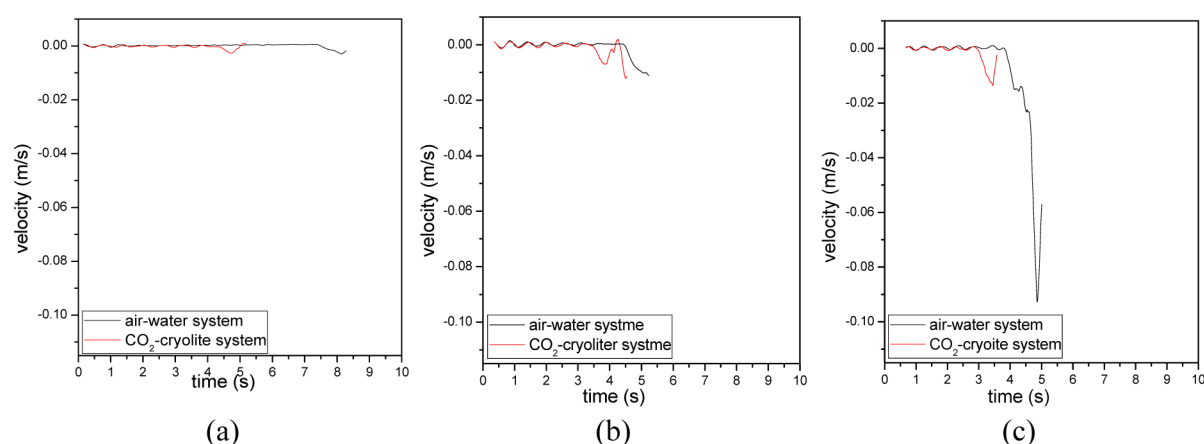


Figure 15. Liquid vertical velocity at point 3 for each case: (a) bubble size 0.0113 m; (b) bubble size 0.0226 m; (c) bubble size 0.0339 m.

and bubble sizes on liquid flow. The velocities in three locations are recorded for this comparison; these are marked in red dots in Figure 1. Point 1 is situated at the middle of the gap between the anode and cathode, right under the anode edge, where the horizontal flow is recorded. Points 2 and 3 share the same vertical height, which is 75 mm below the liquid surface. In the horizontal position, points 2 and 3 are, respectively, located 5 mm from the anode vertical wall and the outside wall. For both points 2 and 3, the vertical velocity is recorded given that the flow is dominant in this direction.

Figure 13 plots the transient horizontal liquid flow velocity at point 1 for each case. A negative value indicates that the liquid flows into ACD. The liquid undergoes a period of quiescence before the arrival of the bubble. Following the passing of the bubble, liquid flows into the ACD, leading to a sharp increase in the liquid flow velocity. Generally, for a given bubble size, the peak velocity is higher in the air–water system than in the CO_2 –cryolite system. The large bubbles lead to a higher liquid flow velocity as more liquid is required to replace the gas. Once the bubble passes the recording point, the liquid flow reduces quickly. Consistent with the bubble sliding velocities (Figure 5), the higher sliding velocity for the CO_2 –cryolite system leads to a shorter period before the liquid flows into the ACD. As shown in Figure 8, there is a strong local recirculation beneath the bubble. In the real system, bubbles are continuously generated, which will lead to the flow of liquid into ACD in a continuous manner and intermittently out of the ACD following the bubble release.

Figure 14 plots the liquid vertical velocity at point 2 for each case. A negative value signifies a downward flow at this point. The passing bubble drives the liquid flowing upward. The magnitude of the bubble induced flow is much stronger in comparison to the flow in the ACD due to the strong buoyancy effect. Again, the flow velocity is higher for a larger bubble. A sharp increase in the flow velocity occurs in the CO_2 –cryolite system. Conversely, in the air–water system, there is a less sharp change of the flow velocity and the flow is moving both up and down. This difference is probably related to the bubble rising behavior (Figure 12), where the bubble in an air–water system has detached from the anode vertical wall.

Figure 15 plots the liquid vertical velocity at point 3 for each case. The liquid moves upward near the anode then changes its direction to flow horizontally, below the free surface, toward the outer wall, and then flows back vertically down beside the outer wall, completing the recirculation. The flow velocity is much lower at point 3 than that in the upward flow close to the anode (Figure 14). Note that the continuous injection of bubbles will likely lead to a high recirculation rate.

It is worth noting that the present work is the initial phase of the investigation into bubble dynamics in aluminum reduction cells, which has been restricted to the behavior of individual bubbles. The bubble flow in an industrial cell will be more complex, as it involves continuous generation of many bubbles which interact and undergo dynamic coalescence and breakup. In addition to the momentum exchange between the bubbles and the liquid, the bubble will induce a strong turbulence into

the liquid which will assist alumina mixing beyond that predicted from single bubble models. Nevertheless, it is only by careful validation of the single bubble models against experimental data that confidence can be gained to approach this larger challenge.

4. CONCLUSION

By applying an advanced VOF-CFD model with mesh adaption technology and using a 2D geometry of part of an aluminum reduction cell as the testing bed, the dynamics of single bubbles have been investigated for both air–water and CO₂–cryolite systems with three bubble sizes.

The results demonstrate that both systems show a similar trend of bubble dynamics: the sliding velocity in the ACD increases as the bubble size increases; large bubbles tend to move with a thick head; smaller bubbles can escape from the anode edge as a whole, while larger bubbles tend to break following the release of part of the bubble.

Quantitatively, there are some differences between the two systems, which can be reflected in the three stages of the bubble release processes:

- Bubble sliding under the anode: For a fixed size, the CO₂–cryolite system leads to a larger bubble sliding velocity and a smaller bubble thickness than in the air–water system.
- Bubble releasing from the anode edge: For an air–water system, the bubble prefers to detach from the anode vertical wall, while, for a CO₂–cryolite system, the bubble would rather attach to the vertical edge of the anode.
- Bubble rising in the side channel: For an air–water system, the bubble continuously detaches from the vertical anode wall, while, for a CO₂–cryolite system, the bubble mainly attaches to the vertical wall. Consequently, the momentum exchanges occur at different regions of the cell channel.

The present detailed investigation of the flow dynamics around the bubble region has indicated that the flow is very complex. The flow velocity in the bubble region exceeds the velocity of the front liquid, leading to a thick head region. The motion of the bubble leads to liquid recirculation moving in the opposite direction to the bubble. The skin friction force at the interface leads to the formation of various local vortices, which will assist in alumina mixing.

The simulation results obtained so far demonstrate the feasibility of the present modeling approach as an effective numerical tool for the comparison of bubble dynamics for different systems and for future proof-of-concept studies. Further work with continuous gas injection of gases is needed to better represent the aluminum smelting process and to build quantitative correlations based on microbubble dynamics for the support of macro process scale simulation.

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Notes

The authors declare no competing financial interest.

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NOMENCLATURE

- C_d = drag coefficient, dimensionless
 d_b = bubble size, m
 $\vec{F}$ = volumetric surface tension force, N m⁻³
 $\vec{g}$ = gravitational acceleration, ms⁻²
 h_{lim} = bubble limiting height, m
 h_{max} = bubble maximum height, m
 L = characteristic length, bubble size in this paper, m
 Mo = Morton number, dimensionless
 P = pressure, Pa
 t = time, s
 $\vec{v}$ = velocity, ms⁻¹

Greek Letters

- α = phase volume fraction, dimensionless
 θ = the liquid–solid contact angle, deg
 μ = viscosity, kg m⁻¹ s⁻¹
 ρ = density, kg m⁻³
 σ = surface tension, N m⁻¹
 φ = inclination angle of the anode bottom surface, deg

Subscripts

- g gas
 l liquid

Superscript

- T the matrix transpose operator

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